TANISHA SRINIVASAN

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EDUCATION

University of Southern California May 2029
Bachelor of Science, Physics & Computer Science
Minor in Computational Neuroscience
Relevant Coursework: Probability Concepts in Engineering, Mechanics and Thermodynamics

University of California, Santa Cruz June 2025 – June 2026
Coursework toward Bachelor of Science, Computer Science 3.90
Relevant Coursework: Computer Systems, C Programming & Assembly Language, Linear Algebra, Vector Calculus, Python

SKILLS

Programming: Python, C, C++, Java, JavaScript
Hardware/Design: Arduino, Raspberry Pi, KiCad, SolidWorks, PCB Design
Tools: Git/GitHub, MATLAB, NumPy, Pandas, OpenCV, TensorFlow/Keras

EXPERIENCE

Product Specialist / Engineer — Dexterra Grip, Project Invent Jan. 2025 – Present
San Jose, CA

- Designed and prototyped a $200 assistive glove using Arduino Nano, force-sensitive resistors, servo actuation, and tendon-driven mechanisms to provide proportional grip assistance for users with limited hand mobility.
- Iterated through multiple actuation concepts and mechanical configurations, refining sensor placement, tendon routing, and servo-driven assistance to improve grip consistency and usability.
- Integrated embedded electronics, custom hardware, and mechanical components into a functional wearable prototype designed around feedback from an end user.
- Selected as a U.S. finalist in the international InventFuture.Global Innovation Challenge and recognized through the U.S. National Innovator Challenge and Project Invent Moonshot Award.

Hardware & Embedded Systems Developer — Tremor-Dampening Device, Pie-Fi Incubator. July 2025 – 2026
Santa Cruz, CA

- Developed a wearable device designed to reduce the effects of hand tremors, contributing to hardware design, prototyping, and embedded-system integration.
- Built and tested sensor- and actuator-based prototypes with teammates while evaluating different approaches to tremor stabilization and mechanical damping.
- Designed mechanical components and integrated embedded hardware into a wearable form factor through iterative prototyping and testing.
- Presented the completed prototype at UC Santa Cruz's startup incubator Demo Day, demonstrating the device to students, mentors, and members of the campus entrepreneurship community.
- Continued exploring improvements to the device's hardware design and portability following the initial prototype.

Developer — API Security Project, Google Developer Group @ UCSC Sept. 2025 – 2026
Santa Cruz, CA

- Collaborated on a Python-based API security system using request validation and replay-attack prevention to protect API credentials.
- Contributed to a proxy-based architecture for authenticating requests and preventing unauthorized reuse of API calls.
- Designed the system with a student development team for integration into production-style software environments.

Research — Exoplanet Detection, Inspirit AI Scholars Aug. 2024 – Oct. 2024
Santa Cruz, CA

- Developed machine-learning models in Python, TensorFlow, and Keras to classify stellar light curves using NASA Kepler data.
- Trained and evaluated convolutional neural networks on a dataset of 100,000+ stellar observations, achieving approximately 87% validation accuracy.
- Presented model methodology and results at a program-wide research symposium.